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CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

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Purpose: The gold standard for naming vertebra at the lumbosacral junction is counting caudally from C2 on whole-spine.
But in practice, a lumbar surgical case is planned on lumbar-only imaging (T12 to S1) without C2, making the gold standard method impossible.
Abdominopelvic computed tomography (CT) holds the lumbar-only imaging span plus the lowest ribs and pelvis, making it a reasonable substitute for the preoperative view.
When a lumbosacral transitional vertebra (LSTV) alters the count, however, the local anatomy is much more ambiguous.
A person who has four rib-free lumbar vertebrae may have an L1 with a lumbar rib or an L5 assimilated to the sacrum, and a person who has six rib-free lumbar vertebra may have an L6, a T12 with aplastic ribs, or a lumbarized S1.
CTSpinoPelvic1K asks whether local morphology can resolve that ambiguity without counting caudally from C2.
Two public label sets already covered this imaging, CTSpine1K's vertebrae and CTPelvic1K's pelvis, on the same colonography patients.
This release joins them in one dataset with 802 CT records, per-level ribs and femora, and lumbar levels anchored on the lowest rib-bearing vertebra and the sacrum.

It also introduces classes for L6, T13 and lumbar ribs, so anomalies are accurately recorded and not forced into whichever normal count they happen to resemble.

Acquisition and Validation Methods: Records pair CTSpine1K and CTPelvic1K labels on each patient's bone-richest acquisition under a VerSe-native scheme.
Validation covered geometric invariants (802/802 pass), rib-vertebra incidence across 5,749 ribs (0.035% offset), and spinopelvic measures matching published values.

Data Format and Usage Notes: NIfTI image/label pairs with patient-grouped LSTV-stratified five-fold splits and a loader; archived at https://doi.org/10.5281/zenodo.22139642.

Potential Applications: Classifying a vertebra from local features; updating cadaveric morphometry; spinopelvic assessment; opportunistic screening; and, absent a public preoperative lumbar cohort, surgical planning research (377 records prone).
Limitations: thoracic ground truth is field-of-view limited; postural angles are supine; no held-out test set; ribs are triaged-review pseudolabels; Castellvi grades are two-reader consensus on 33 records.

The spine typically has seven cervical, twelve thoracic, and five lumbar vertebrae.
A vertebra is classified by where it falls in that sequence.
Transitional variants sit at the two ends of the lumbar column, with the thoracolumbar transitional vertebra (TLTV) above and the lumbosacral transitional vertebra (LSTV) below.
Both transitional variants disturb identification by changing what the border vertebra and its ribs look like.
For example, a thoracolumbar border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral one may be partly or completely assimilated to the sacrum (sacralization), with an anteriorly wedged body, a reduced disc beneath it and hypoplastic facet joints, or separated from it (lumbarization), with a squared body, lumbar-type facets and a full-height disc[1].
In addition, they determine whether there are eleven or thirteen thoracic vertebra and four or six lumbar vertebra.

Together, these transitional variants they make it difficult to definitively classify the vertebra at these borders.
Each of those readings yield the same count (Fig.
2), and

the morphology that would separate them is not always decisive to human readers.
Whether it differs enough for a trained classifier to separate them, or at least to assign each reading a probable classification, is the question this release aims to answer.
The gold standard for numeration in transitional anatomy is whole-spine imaging, counting caudally from C2[2], which determines how many vertebrae and how many rib pairs the column holds.
No spine-limited acquisition can supply either count.
LSTV is common, reported at 10–29% in the general population[7] and in 16.3% of a whole-spine CT series,[3], with potential for wrong-level surgery being its most serious consequence.

Existing public collections cannot address this, and not for lack of size (Table I).
Prior spine datasets omit the pelvis and pelvic datasets do not number the vertebrae.
TotalSegmentator[4] has no class for a sixth lumbar vertebra or for a rib on a lumbar vertebra, and VerSe[5], the one collection with an L6 class, stops at the sacrum and carries no ribs.

As such, none of these collections can record a spine with six lumbar vertebra together with

both of its borders, and a segmenter trained on them must shift the whole column by a level or absorb a vertebra into its neighbor when it encounters an enumeration anomaly.
This problem has now been measured on CT, where prior labeling methods assigned every vertebra correctly in 77% of subjects and an anomaly-aware extension of SPINEPS^{6} raised that to 99%^{7}.
Its weights label L6 and T13 on CT, but its 1,536-scan CT cohort is in-house and unreleased, so what exists is a capability, not a dataset.

When considering the intraoperative side, LevelCheck registers the intraoperative radiograph to the preoperative CT and projects the CT's vertebral labels onto it^{8-10}.
Those labels, however, are placed by hand and verified by the surgeon^{8}, so the method assumes a correctly labeled CT and does not supply one itself.
As such, a dataset carrying the anomaly classes is vital for automating that step.

CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and classifies each structure an enumeration anomaly produces, including a sixth lumbar vertebra and lumbar ribs.
A scheme without those classes is unable to adequately describe anatomic anomalies.

The project began with the question whether a vertebra can be named from the anatomy on the images a lumbar surgical case is actually planned on.
By guideline those images are lumbar-only^{11-14} and whole-spine coverage is reserved for trauma with an identified injury^{15,16} and for deformity radiographs.
An abdominopelvic CT runs from the diaphragm to the pelvic floor, so it captures that span with the lowest ribs and the whole pelvis, while lacking C2.
Every record comes from one prospective trial protocol, ACRIN 6664^{17,18}, which scanned 802 patients aged 50 and over, supine and prone, on multi-detector CT at 15 centers, on five scanner vendors and nine models, with manufacturer, model and reconstruction kernel recorded per record.
To our knowledge, it is the largest spine-and-pelvis annotated CT cohort acquired under a single protocol.
The comparators, by contrast, pool collections, are multi-site by design, or are routine clinical scans under many protocols (Table I).
With the protocol for this study fixed, scanner effects on a model become measurable rather than confounded.
The downside the restriction to an abdominal field of view, in which only the lowest thoracic levels are present (Sec.
V.

The gap was evident.
CTPelvic1K^{19} and CTSpine1K^{20}, each annotated the COLONOG collection under radiologist supervision for the same patients, with one annotating the pelvis and the other annotating the vertebrae.
The datasets had remained separate until now, despite the fact that no new imaging or radiologist was needed.
It was not a file merge, because each annotation had to be traced back to the series it was drawn on (Sec.
II A.
Once completed, that merge yielded a combined spine-and-pelvis frame for 802 patients, and the bones from both sets had ribs, femora and hardware that

could be annotated against it rather than from scratch.

The main limitation is that no scan in this dataset contains C2, so the gold standard conventional count cannot be performed.
This is a property of abdominal imaging rather than of the annotation itself.
A thirteenth thoracic vertebra and an L1 with a lumbar rib are different phenotypes, distinguished as separate subtypes in cadaveric classifications of the thoracolumbar junction^{21,22}.
The first is an additional rib-bearing vertebra positioned above five lumbar vertebrae, so the column has 25 presacral vertebrae.
The second is a rudimentary rib on a lumbar-type L1 in a column of with the standard 24 presacral vertebra.
Neither is separable by counting from last rib-bearing vertebra or from the sacrum because four rib-free vertebrae can fit a lumbar rib, a cranially shifted T13 or an L5 assimilated to the sacrum, and six rib-free vertebra can fit an aplastic twelfth rib or a sixth lumbar vertebra.
A stump rib is a hypoplastic rib that is a key indicator of a thoracolumbar transitional vertebra.
On whole-spine CT scans, rib anomalies travel with the lumbosacral border and hypoplastic twelfth ribs occur with sacralization and lumbar ribs with lumbarization.^{3}

In this release, every vertebra carries the identifier its radiologist-sourced annotation assigned as the ground truth, corrected where the source was wrong.
In addition, every released measurement is expressed against the two landmarks present in essentially every abdominal CT scan: the sacrum inferiorly and the lowest rib-bearing vertebra superiorly.
This makes the measurements positional rather than nominal, with a vertebra being classified by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its length as a fraction of the rib above it, and a transverse process by its span and its distance from the ala. None of these values shift with the name a reader gives the junction, meaning that cases that disagree about the name remain comparable.

But the vertebrae themselves are known to differ.
Level-specific morphometry is long established in the literature: pedicle width and height change systematically from the thoracic to the lumbar spine,^{23} vertebral body, endplate and canal dimensions differ by level,^{24-26} and the transverse process and the iliolumbar ligament mark the last lumbar vertebra independently of any count.^{1,27} If those differences hold at the boundary, the phenotype is able to be determined locally, and on this dataset it is.
Separating T12 from L1 on shape alone gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae.
This was calculated using a logistic regression on five released per-level measures and six ratios between them.
Each measure was divided by the patient's median across their levels, so that size, which separates thoracic from lumbar trivially and says nothing at the junction, is removed.
Folds are grouped by case and the curve is taken on pooled out-of-fold scores

FIG.
1 How the dataset was built.
n counts scans, or corrections, at each step.

(testmorphometricseparability.py in the released code).
Schinz et al.
reach the same conclusion from the other side: on 1,242 whole-thoracolumbar CTs a shape-based labeling of the junction matched nerve morphology in every case, against 92.6–97.2% for counting- and rib-based rules.
28

II A.
Sources, and why a crosswalk was necessary

CTSpine1K's 1,005 volumes and CTPelvic1K's 1,184 volumes consist of four and seven collections, respectively, of CT scans obtained for different indications.
Of these, only the Cancer Imaging Archive (TCIA) CT colonography collection (COLONOG)\(^{17,18}\) is shared, and it is the largest in both, consisting of a subset of the images collected through a prospective screening trial of asymptomatic patients fifty years and older under a uniform protocol.
Inspection of the code released with CT-Spine1K indicated that the volume for each annotation was arbitrarily chosen, via os.walk in DICOM directo-

ries with twelve or more files, from the available acquisitions for each patient, while no code was released with CTPelvic1K.
This arbitrariness is demonstrated by the heterogeneity of the annotated series in Table II.

Because every patient in COLONOG was scanned in more than one position, at the bare minimum prone and supine, and the SeriesInstanceUID from the original DICOM header has no counterpart in NIfTI and is lost on conversion (Table III), determining which series in TCIA each annotation was drawn on is nontrivial.
CT-Spine1K released paired CT volumes with each label and Path.csv, which is claimed to map to the TCIA directory each was converted from.
However, that file doesn't include a SeriesInstanceUID, lacks a position for 217 of the 782 annotated records, and indicates a different acquisition than the label-bone overlap maximization described in this work for seven (Table IV).

CTPelvic1K released no paired imaging with their labels, encoding the mapping to the TCIA volumes in the mask filenames instead, as dataset2PatientIDSeriesNumberundocumented[tag] mask4label.nii.gz.
The first is the TCIA subject identifier (...50.4.0001).
The second is the DICOM SeriesNumber.
Using these two keys 702 of 713 masks are in alignment with the volumes chosen by bone overlap maximization.
The third field is undocumented, and the fourth is an optional suffix on 32 of 714 masks, describing phenotype, annotation difficulty, and image condition in one string (Table V).

listing of the source directory each was converted from; that listing carries no series identifier, names no position for 250 of its 815 rows, and contradicts itself on a further 70.
CTPelvic1K released labels alone, leaving the user to obtain and pair the imaging themselves.

Turning a patient from supine to prone alters lumbar lordosis and the alignment of each segment against the next, so outlines drawn on one series are the wrong shape for the other and no rigid realignment recovers them.

The crosswalk (Fig.
1) therefore resolves each annotation to a patient and then, within that patient's volumes, to the series whose bone agrees with it, with candidates thresholded at bone attenuation and scored by overlap with the mask. This step mattered for 351 patients, nearly half the cohort, whose two collections had labeled different acquisitions of the same person; those records are released but held out of any analysis assuming a single posture, since two labeled postures of one patient are a resource rather than a defect.

Densification, and why it is asymmetric

A corpus in which half the records lack a pelvis is not a spinopelvic dataset, so the missing structures were pseudolabeled, asymmetrically (Fig.
Pelvis onto spine-only records is the safe direction, since the sacrum and innominate bones carry no enumeration to get wrong, and quality is reported as held-out performance on the pelvic_only records, withheld from training for that pur-

TABLE I Public CT collections at the lumbosacral junction.
Black, class present; outlined, graded only to exclude it \(^{29} \) .

| Collection | Scans | Count | L6 | Sacrum | S1 | Pelvis | Ribs | L. rib | Femora | Grade |

CTSpine1K 20 1 005

CTPelvic1K 19 1 184

VerSe 5 374

RibSeg v2 30 660

TotalSegmentator 4 1 204

CTSpinoPelvic1K 802

TABLE II Properties of the series in the CTSpinoPelvic1K dataset; three annotations sit on a decubitus series and four on a 5 mm reconstruction.

| Position | n | Slice thickness | n |

| Supine | 412 | 0.625 mm | 3 |

| Prone | 367 | 0.9 mm | 1 |

| Decubitus right | 2 | 1.0 mm | 546 |

| Decubitus left | 1 | 1.25 mm | 228 |

| | | 5.0 mm | 4 |

TABLE III What identifies a CT series, and what survives conversion to NIfTI; the NIfTI-1 header has no field for patient, study or series.

| DICOM attribute | Identifies | In NIfTI |

| PatientID (0010,0020) | the person | no |

| StudyInstanceUID (0020,000D) | one exam visit | no |

| SeriesInstanceUID (0020,000E) | one acquisition | no |

SeriesDescription 0008 103E Colosupine 1 0 B30f no

| PatientPosition (0018,5100) | prone or supine | no |

| ConvolutionKernel (0018,1210) | reconstruction kernel | no |

| SliceThickness (0018,0050) | 1.0 mm | via pixdim |

TABLE IV How each source's mapping resolves against the series selected here by bone coverage.

| | n |

| CTSpine1K, Path.csv, annotated records | 782 |

| names no position, so cannot be checked | 217 |

| names a position, and it agrees | 558 |

| indicates a different acquisition | 7 |

| CTPelvic1K, filename fields, pelvic masks | 713 |

| SeriesNumber resolves to the selected series | 702 |

| SeriesNumber shared by two series | 5 |

| SeriesNumber resolves to a different series | 6 |

| third field resolves to no series at all | 713 |

TABLE V The optional tag in CTPelvic1K's mask filenames, on 32 of 714; one free-text field carrying three unrelated kinds of information.

| Tag | Records | n |

absent 682

| sacralization | phenotype | 7 |

| semisacralization | phenotype | 2 |

| hard_sacralization | phenotype, difficulty | 1 |

| hard | annotation difficulty | 8 |

| veryhard | annotation difficulty | 2 |

| lowdose | image condition | 6 |

| metal | image condition | 1 |

| crop | image condition | 1 |

| IntestinalCalculus | image condition | 1 |

| dgjoint | unstated | 2 |

| ydjoint | unstated | 1 |

pose.
The reverse direction carries the whole problem this dataset addresses, because a pseudolabeled spine must commit to a count, and on a transitional vertebra it commits to the commoner reading; those 20 records were corrected by hand.

No public rib annotation covers this cohort, and the two available automatic sources \(^{4,31} \) fail in complementary ways.

Möller's binary rib network \(^{31} \) reports high accuracy, and its authors show that stump ribs can be classified from a partial field of view; that result concerns the morphological features computed on a rib once segmented. Here, though, the segmentation itself failed on ribs that were partially outside the field of view.
A rib crossing the boundary of an abdominal acquisition was liable to be missed altogether rather than segmented to the edge.

In a colonography cohort the lower ribs are routinely clipped, and those happen to be the ribs the count depends on.

TotalSegmentator \(^{4} \) , by contrast, does segment clipped ribs and supplies the per-rib numbering that a binary network cannot.
Its characteristic failure is at the other end of the bone, where the most medial portion of the rib (roughly the last tenth to sixth approaching the costovertebral joint) is frequently absent.
A rib truncated at its medial end never reaches the vertebra it articulates with, so the rib-vertebra incidence check (the quality-control gate on the rib class labels, which matches each rib to the vertebra its number implies) has nothing to test.

Given these complementary gaps, the two approaches were combined (Fig.
1), TotalSegmentator supplying the numbering and the medial-clipped cases, Möller the detection where its output is more complete.

The result is a pseudolabel whose review was triaged rather than exhaustive.
A label-based rule flagged every record in which one rib bone carried more than one identifier, or in which a rib failed to reach the vertebra it articulates with, since a single bone carrying two numbers is a numbering error whatever the anatomy.
The rule selected 152 records; 148 were reviewed and corrected, and four were not and are identified in the release.
Reviewers were medical students working through a student research consortium \(^{32} \) , trained against a written labeling protocol, with every correction attributed to its annotator.
Records passing the rule were not individually inspected; the residual rib-vertebra offsets reported in Sec.
II.F are the independent check on what the rule missed.

Label scheme and counting anchors

The scheme is VerSe-native.
Vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–

L6 = 20–25, sacrum 26, coccyx 27, T13 = 28), and every non-VerSe structure takes a fixed identifier above that range, hips (30–31), femora (32–33), thirteen ribs per side (34–46 left, 47–59 right), lumbar ribs (60–61) and surgical hardware (62–68).
Identifier 29 is retired and unused, for the reason given below.
Rib 13 is the true rib of a T13 and is empty in this release, so a thirteenth thoracic vertebra and a lumbar rib are recorded as the distinct phenotypes they are.

Two anchors give every record the same positional frame, and Fig.
2 shows both, the lowest rib-bearing vertebra above and the sacrum below.
The sacrum's superior end-plate is the landmark sacral slope and pelvic incidence are measured from, and with both brackets present L5 and L6 are distinguishable where a spine-limited view alone would not be.

A separate S1 label was withdrawn, and why.
Earlier releases carved the cranial portion of the sacrum as its own S1 class, on an automatic estimate of the S1–S2 boundary.
Measured against the anatomy on all 802 records, that estimate is unreliable: S1's craniocaudal extent is a median 57.6 mm where a first sacral segment is near 30, only 26% of records fall in a defensible range, and in 222 the plane takes more than half the sacrum.
The failures are not merely low cuts.
Where the fit degenerates the plane comes out nearly parallel to the sacral long axis and shaves the ventral cortex, so the retained sacrum is left without its anterior wall — wrong for every use of the label, not only for the end-plate.
Rendering one such record is what showed this; no summary statistic did.

The consequence was confined to the parameters fitted to that surface.
Splitting the cohort on whether the carve was anatomically sound separates them cleanly: the 580 sound records give a pelvic incidence of 52.7° and a pelvic tilt of 19.6°, against Hasegawa's supine-CT 53.4 and 19.2, while the 221 degenerate ones give 68.5 and 33.6, and the end-plate gate described in Sec.
II.E fires on 96% of the second group and 4% of the first.
The measurement was never at fault; the surface it was given was.
So S1 is merged back into the sacrum, restoring the bone the source annotation actually delineated, and the end-plate is fitted to the whole sacrum instead.

VerSe[5] carries L6, whose identifier this release keeps, so its L6 is VerSe's.

A rib on a lumbar body receives its own class, and this is the one class here with no published counterpart.
A lumbar rib and a stump rib are the same object under two counts, so a scheme that numbers every rib 1–12 leaves the annotator with two bad options: either calling it rib 12, which asserts that the vertebra beneath it is thoracic, the very question at issue, or discarding it.
TotalSegmentator carries ribs 1–12 per side and no such class, and VerSe's T13 is a vertebra rather than a rib.
This scheme records both: VerSe's T13 (28) with its rib pair (46, 59) for a thoracic-type thirteenth vertebra, and 60/61 for a lumbar-type body bearing a rudimentary rib.
No record here carries a T13; the sixteen lumbar-rib

cases were read as lumbar-type bodies, and where a border vertebra could not be settled by its readers the label stands as read.

Figure 2 shows why the interval is the primitive and the label is derived.
Among the records with four rib-free vertebrae, some reach that count from above, through a rib on L1, and some from below, through a lumbosacral segment fused to the sacrum, in Fig.
2(c) together with stump twelfth ribs; the count alone cannot tell them apart, and the release can because ribs, rib lengths and the lumbosacral junction are all labeled.

Sixteen records carry a lumbar rib, thirteen bilateral and three unilateral, all on the right, too few to speak to side preference.

Eighteen records carry an L6.

Seventeen of them carry a consensus Castellvi grade; the eighteenth was not graded.
The source transitional label (pelvic where CT-Pelvic1K flagged one, otherwise the lumbar count) reads lumbarization in fourteen, sacralization in two and semi-sacralization in one, and is unremarkable in the ungraded record.
A spine can carry six lumbar-type vertebrae and still be read as a sacralization, depending on where the reader started the count.

Both counts, and the manifest fields that report them (hasl6, haslumbarrib, nlumbarlabels, lumbarrib_side), are computed from the released label volumes by counting identifiers 25 and 60–61.

Computational tools

Access.
Every step is performed by open-source code archived with the dataset, together with the reference loader, the label scheme and the quality-control scripts that generated the tables in this manuscript.

Versions and parameters.
Segmentation of femora, the sacral sub-division and the per-level rib numbering used TotalSegmentator[4] (total task, release 2.x) on a single graphics processing unit.
The binary rib network is Möller's[31] released weights, applied through the nnU-Net v2 framework[33].
The pelvic completion for records whose pelvis was absent used a five-fold nnU-Net v2 ensemble applied out-of-fold, so that no record is completed by a model that has seen it.
The five fold checkpoints, with their nnU-Net plans and dataset descriptor, are released alongside the data (the repository URL identifies the authors; it is on the title page and will be restored here for the camera-ready version).
Image handling throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is written, so every label shares its image's grid and affine exactly.

Measurement.
Dimensions and spinopelvic parameters come from code released with the dataset.
A sacral plate whose fitted normal lies more than 60° off the cranial axis is rejected and the case reported missing, since such a fit has found the near-vertical anterior face of the promontory, not the endplate.
Gating on the geometry rather than on the value leaves unusual patients in, an open-source measurement toolkit is released separately (the

FIG.
2 The two anchors, from the released labels, aligned on the sacrum.
Red, lowest rib-bearing vertebra and its rib; blue, the sacrum; yellow, the rib-free vertebrae between.
Cases 0704, 0428, 0094 and 0005, chosen by rule as the most ordinary member of each phenotype (scripts/selectanchorcases.py).
(b) and (c) both count four.
In (b) a long rib sits on L1; in (c) the twelfth ribs are stumps and the segment below the fourth lumbar body is fused to the sacrum (Castellvi IIIb).

URL identifies the authors and is on the title page) as a reusable implementation measuring in each vertebra's own frame; no open-source implementation computing these parameters from CT segmentations was found elsewhere.

Generative artificial intelligence.
A large language model (Claude, Anthropic) assisted with writing the analysis and figure code and manuscript refinement after initial drafting.
It was not used to generate, annotate or interpret imaging data; every number reported here is computed by the released code from the released volumes.
All content was independently reviewed, verified, and interpreted by the authors.

Validation has three layers.
Geometry is checked first, because a measurement taken on a corpus with a geometric fault still returns a plausible-looking number.

Geometric invariants.
Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left-right axis read from the affine rather than assumed.

The check compares each sided pair separately.
Testing the ribs alone passed all 802 records, four of which carried a left_hip label on the patient's right side; pooling every sided structure into one test still missed one of those four records, because a large correctly-sided structure masks a smaller transposed one.
A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

Rib-vertebra incidence.
Each rib is matched to its nearest vertebra and compared against the vertebra its number implies.
Of 11,548 ribs, 5,788 are not evaluable because the expected vertebra lies outside the field of view and 11 have no vertebra within the anchor distance; of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is misnumbered.
The denominator is stated because quoting the two offsets against all 11,548 ribs would halve the rate by counting

ribs the check never examined.
Both offsets are field-of-view truncations, one at +1 level and one at -1.

The same check carries an invariant that reads as a null result.
Of the 5,749 evaluable ribs, zero are assigned to a lumbar vertebra, not because no lumbar ribs exist, since 16 records carry one, but because such a rib takes class 60 or 61 and never enters the numbered set the test examines.
A numbered rib landing on a lumbar vertebra would be a counting error of exactly the kind this dataset exists to expose.

Agreement with published values.

Pelvic incidence is the strongest of these checks because it is a morphological property rather than a posture; it agrees to within half a degree with the automated supine CT series of Veilleux et al.[34] and sits below the standing figure, the direction reported for subjects imaged both ways.[35] Sacral slope and pelvic tilt are postural, and read three degrees below and four above Veilleux's asymptomatic subjects; a second supine CT series reports 53.4, 34.1 and 19.2°, within a degree of this cohort on all three.[36] Recumbency runs the wrong way to explain the gap, raising slope rather than lowering it, so the residual tracks cohort and not construction.

Castellvi typing

Every record whose vertebral count departs from five carries a Castellvi grade[37] in the released manifest, 33 cases typed I–IV with the a/b unilateral–bilateral qualifier.
The grade and the count are different axes.
Grade IIIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a normal count of five.

Two radiology resident physicians graded every case independently and resolved their six disagreements by consensus; both reads and the consensus are in the manifest.
The distinction the readers disagreed on, type II (articulation) against type III (bony fusion)[1], is the one that matters clinically, and four of the six disagreements were III against II.

7

TABLE VI Spinopelvic measures, mean ± SD over the 756 of 802 records passing the geometric gates, against two supine CT series that report means.34,36 Standing values are 55.0, 41.0, 13.0 and 60.0°.38 The end-plate is fitted to the whole sacrum; the gate rejects 14 records where the previous carved-S1 fit lost 239.
Pelvic incidence and tilt are right-skewed (0.25 and 0.33), so their medians—51.5 and 15.0°—sit below these means and the modal peak in Fig.
4 sits below both.

| Measure | This dataset (supine, n = 756) | Veilleux (n = 200) | Hasegawa (n = 24) |

Pelvic incidence 52 1 11 1deg 52 1 53 4

Sacral slope 36 6 10 6deg 36 5 34 1

Pelvic tilt 15 5 7 9deg 15 6 19 2

Lumbar lordosis 51 7 13 3deg

PI LL mismatch 0 3 10 3deg

Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here because an iatrogenic fusion is indistinguishable from a congenital one to a distance measurement.

A cage-bridged interspace reads as “no gap” exactly as a fused transitional vertebra does.

II H 0 a.
Detection over-calls by a factor of five.
A 2500 HU threshold inside a shell around the labeled skeleton flags 52 of the 802 records.
All 52 were reviewed manually, confirming 11 and rejecting 41 (79%) as contrast, calcification and reconstruction artefact.
Attenuation does not separate the two groups (both saturate); site and volume do, every confirmed implant at 2,586 mm³ or above and every rejection at 1,768 mm³ or below.

II H 0 b.
The subtype block had to be extended.
Identifiers 62–65 name spinal instrumentation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds, so 66 arthroplasty, 67 sacroiliac screw and 68 osteosynthesis were added.
Osteosynthesis holds parts of one bone together where arthroplasty replaces a joint, and a shape rule calls a femoral stem a rod.
The confirmed cases are eight arthroplasties, one osteosynthesis, one sacroiliac fixation and one pair of interbody cages (Fig.
3.

Metal outranks bone, and that is a subtraction.
In every confirmed case the implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than adding them.
In eight of these cases the femoral head that pelvic incidence and tilt are measured from is an implant.

Each record is a gzipped NIfTI image/label pair sharing an identical 4 × 4 affine, canonicalized to PIR, requiring no resampling.
Masks are NIfTI rather than DICOM, which is what volumetric segmentation tooling expects.

Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the data as splits.5fold.json.
The five validation folds partition all 802 records, so the release provides cross-validation and not a held-out test set; users reporting a single held-out number should carve one out and state which records it holds.

Stratification is on the transitional subtype rather than a binary LSTV flag.
Across all 802 records the source transitional label reads lumbarization in 14, sacralization in 17 and semi-sacralization in two; separately, 18 records

carry a sixth lumbar-type vertebra, nine carry only four lumbar labels, and 33 carry a consensus Castellvi grade.
A record called sacralization by its four lumbar labels enters the same stratum as one called sacralization by the source label, so the two rare strata hold 15 records each and every validation fold carries three of each.
That is the most stratification can guarantee, and it is why a fold-level metric on the rare classes carries an error bar far wider than its own decimal places.

The archive of record is Zenodo (https://doi.org/10.5281/zenodo.22139642).

Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

The cohort is a colorectal cancer screening population of 802 records: 393 recorded female and 345 male, 11 carrying DICOM's other value and 53 no sex field, with age present for 709 (median 59, range 50–89).
All 802 are counted in totals; the eleven outside the female/male dichotomy are excluded from sex-stratified measures.

Count-free measures.
Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which is present does not by itself determine what to call it.
The lowest-rib length ratio is bimodal, and its lower mode, below 0.33 of the rib above, marks a stump twelfth rib in 98 of the 789 records with a measurable pair (12.4%, against 12.6% on whole-spine CT[3]); 16 records carry a lumbar rib (2.0%, against 1.5%).
The co-occurrence reported there is testable here only where the junction was read, and only within those 789: they hold 14 of the 17 source-labeled sacralizations, and stump ribs accompany four of those 14 and 94 of the other 775 (odds ratio 2.9, p = 0.08, Fisher's exact test).
No lumbarization carries a lumbar rib.

Spinopelvic measures (Fig.
Pelvic incidence sits on the standing reference, as a morphological property should; slope and tilt sit below and above the published CT values, which Table VI places against two series.

Opportunistic measures.
Every scan carries a vertebral bone density measurement at no additional dose.39 L1 trabecular attenuation is reported at the published standard site.

FIG.
3 Instrumentation in the release, one exemplar each, drawn through bone.
a Hip arthroplasty 66 n 8. b Femoral neck screws 68 n 1. c Sacroiliac screws 67 n 1. d Interbody cages 63 n 1. Bar 5 cm.

TABLE VII Data types and metadata and the records populating each from the released manifest.

Field Records Available

Identifiers and paths configuration match type image label alignment check 802 100 0

Patient position tube potential section thickness kernel 802 100 0

Transitional label and class annotation origin instrumentation flags 802 100 0

Spine series identifier bone fraction check 782 97 5

Scanner manufacturer and model 768 95 8

Sex 749 93 4

Age 709 88 4

Pelvic series identifier 713 88 9

Castellvi grade consensus of two readers 33 4 1

standing radiograph published CT

FIG.
4 Spinopelvic measures against standing reference bands mean SD 38.

The principal use is the spinopelvic interface.
Spine and pelvis carry radiologist sourced annotation on one coordinate frame in 802 records making sacral morphology measurable against the lumbar column rather than in isolation among them the sacral endplate and its slope the alar corridors constraining S2 alar iliac and iliac screw trajectories and the L5 S1 geometry a lumbosacral fusion must cross.
Pelvic incidence is derived from the segmented sacrum and femoral heads so it is reproducible from the labels themselves.

The transitional layer serves that use rather than competing with it since no trajectory can be planned across a junction whose segments cannot be named.

A proxy for the preoperative view.
No public preoperative lumbar CT cohort exists that we know of so this release is the closest available stand in for the view a lumbar case is planned on it holds the T12 to S1 span with the lowest ribs and the pelvis 377 of its 802 records were acquired prone the position of posterior lumbar surgery and it carries classes for the anomalies that make level identification fail.
It is a screening population at colonography dose not a surgical series so it supports method

development for level identification and instrumentation geometry rather than validation of a planning decision.

Naming a level from its own geometry.
Wrong level spine surgery runs at roughly one in 3 110 spinal procedures and among the predominant contributing factors is the transitional anatomy this cohort was assembled around.
40 41 Per level body height canal width endplate width transverse process span and wedge ratio are released across all 802 records and because the labels were assigned against the twelfth rib rather than by enumeration a model trained on them is not learning the convention that fails.

Reference morphometry with its spread.

The values a surgeon consults for endplate and canal dimensions and pedicle width come from cadaveric series of ten to thirty specimens 24 26 redrawn in the textbooks as one curve per level 42 and a standard error describes the mean not the spread of individuals.
The same measures are given here from 802 records with the distribution attached Fig.
5 reproducing the classical pedicle widening into the lower lumbar spine.
Both layers of that figure are population intervals rather than confidence intervals and that dis

9

Panjabi cadaveric mean n 12 per level 1 96 SD 95 of his specimens

FIG.
5 Morphometry by level T11 L5 median interquartile range 5th 95th percentile n at right.
a End plate width EPWu.
b Canal width SCW and depth SCD.
c Pedicle width PDW both sides averaged.
L5 is withheld in a and c see text.
Reference is Panjabi s cadaveric series 24 25 keyed in a its SD recovered from the reported SEM T11 T12 EPWu is digitised from his Figure 4A.

FIG.
6 Records carrying each vertebral level cranial to caudal.
Counts per identifier in Table S1.

tinction is what makes their widths comparable a confidence interval bounds a parameter a population interval bounds individuals 43 and at n 12 the two differ by sqrt 12.

Patient specific models.
Patient specific finite element modeling for alignment planning and implant design 44 needs spine pelvis and femoral heads in one frame which is this release s construction and 351 patients give a within patient postural change to check a prediction against.

Limitations coverage and cohort.
Thoracic ground truth is field of view limited Fig.
6 and does not reach T1.
Postural angles are supine pelvic incidence is not postural but it is modality sensitive reading below radiography for the same subjects.
35 End plate and pedicle width are withheld at L5 where the transverse processes arise in front of the canal wall the body is cut at.
Those two and the spinopelvic angles which rest on the sacral end plate are the provisional part of this release the planned revision takes body and sacral plate from a substructure label and the version DOI carries it.
The cohort is one protocol of a colorectal screening population aged 50 and over so its distributions should not be read as a surgical one.

Label strength varies by structure and the release does not average over that.

Vertebral labels derive from radiologist supervised source annotations corrected where those sources were wrong pelvic labels on records that lacked one are pseudolabeled.
The rib layer is a pseudolabel whose review was triaged so its guarantee is the absence of the failure modes that rule detects.
Transitional labels come from two independent sources and are not uniformly adjudicated which is why the measures reported here are count free the Castellvi grades are a consensus of two radiology resident physicians.
The sacrum is the source annotation s own outer boundary

throughout no sub division of it is asserted the separate S1 label of earlier releases having been withdrawn as unreliable.

Absent structures and no classifier at the lower border.
Sacrum hips and femora are present in all 802 records one lacks T12 outside the field of view.
Nine carry no L5 identifier because their source annotation counted four lumbar vertebrae all nine are Castellvi IIIb the fused segment is delineated with the sacrum as the caudal anchor and the manifest names them.
No shape based classifier is attempted at the lumbosacral junction 15 records in each rare stratum leave three per fold.
The 0 990 area under the curve above is the thoracolumbar boundary not this one.

Comparison with related material.
VerSe 5 29 comes closest in intent to this dataset but lacks a sacral mask for the structures described by Castellvi grades.
VerSe 5 29 segments these structures as L6 or as sacrum and segments neither the pelvis nor femora a spinopelvic measurement needs nor prone acquisitions.
What this release adds over TotalSegmentator 4 is the classes for anatomical anomalies not necessarily more anatomy per scan.

Future directions.
Applying the released weights to VerSe would add the sacral anchor and the anomaly classes to a bone kernel collection.
A Castellvi read of the whole cohort would test the co occurrence of rib and lumbosacral anomalies at full power and supply the labels a lumbosacral classifier needs.
A class label from substructure segmentation 6 would measure the body without the canal cut that withholds L5 here.

CTSpinoPelvic1K places radiologist derived spine pelvis and pseudolabeled per level ribs and femora on one validated coordinate frame in 802 records and creates a classification for the anatomical anomalies that make lumbar numbering ambiguous.
Because both counting anchors are explicit a level can be identified from a field of view that cannot support the conventional count downward from C2.

The release described here is v10 permanently archived at https doi org 10 5281 zenodo 22642578 the concept DOI https doi org 10 5281 zenodo 22139642 resolves to the current version.

The archive carries the loader the label scheme and the quality control tables under an open source licence.
The repository additionally carries the extraction and figure code the reference table with a citation on every row the values plotted in Fig.
5 and a recipe for running them in order.
The repository URL identifies the authors it is on the title page and will be restored here for the camera ready version.

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ETHICS

This work uses publicly available de identified imaging and annotations derived from it.
The authors institutional review board determined that it is not human participant research under 45 CFR 46 and requires no oversight.

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High Human Impact High AI Impact

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 17 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scene Writing Reports as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our Writing Reports through Google Docs). Check out our list of several recommendations on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.